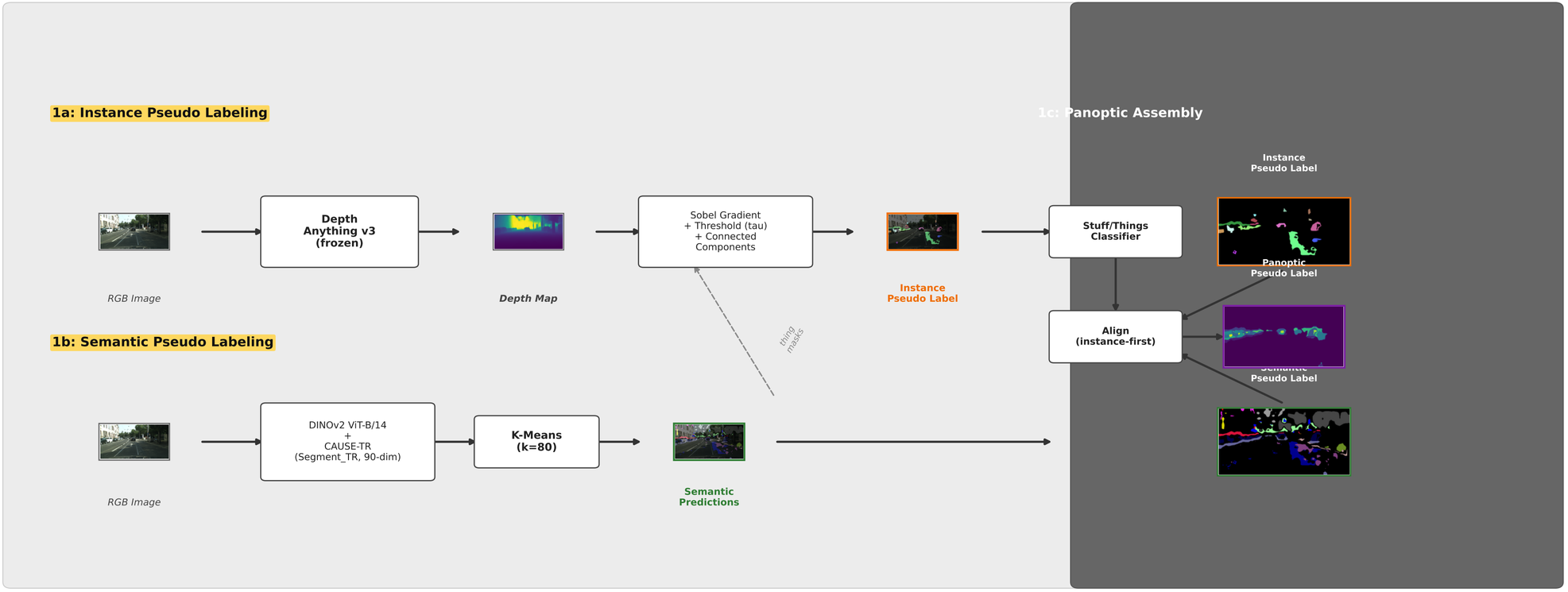

Figure 3. Stage 1: Pseudo-label generation. Instance pseudo labeling (1a) applies Sobel gradient thresholding on monocular depth (DAv3) to split thing-class regions into individual instances via connected components. Semantic pseudo labeling (1b) clusters CAUSE-TR features with k=80 K-means overclustering. Panoptic assembly (1c) aligns the two signals using instance-first merging with a stuff/things classifier based on depth-split ratio.